Exercises: Forces and momentum

IBDP Physics SL · Grade 11 · A.2 Forces and momentum · 24 calculation exercises in three IB levels

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Name: Class: Date:

How to use. Show your working: equation → substitution with units → answer with unit and sensible significant figures. Use g = 9.81 m s^{-2}. Level 1 = one step (Paper 1 style), Level 2 = multi-step, Level 3 = structured exam questions with marks.
Check yourself online (instant marking, hints and worked solutions, new numbers to practise again):
https://school.taneski.com/physics/dp/ex-forces-momentum/

Level 1 · Paper 1 style — one step

8 exercises · 17 marks · show your working

1.1 Car suspension springs A.2 · Types of forces · [2 marks]

A car of mass 1300 kg rests on four identical springs that share its weight equally. Each spring is compressed by 4.0 cm. Calculate the spring constant of one spring.

Answer: _____ N m^{-1}

1.2 Pushing a heavy box A.2 · Types of forces · [2 marks]

A 37 kg box rests on a level floor. The coefficient of static friction between the box and the floor is 0.55. Calculate the smallest horizontal push that will start the box moving.

Answer: _____ N

1.3 Rock under water A.2 · Types of forces · [2 marks]

A rock of volume 500 cm^3 is fully submerged in a lake (ρ_water = 1000 kg m^{-3}). Calculate the buoyant force on the rock.

Answer: _____ N

1.4 Car acceleration A.2 · Newton's laws · [2 marks]

A car of mass 1200 kg has a driving force of 2600 N. The total resistive force is 1000 N. Calculate the acceleration of the car.

Answer: _____ m s^{-2}

1.5 Delivery drone A.2 · Newton's laws · [3 marks]

A delivery drone of mass 1.4 kg is lifted by the upward thrust of its rotors. Air resistance is negligible.

(a) Calculate the thrust when the drone hovers. [1]

(a) Answer: _____ N

(b) Calculate the thrust needed to accelerate upwards at 3.0 m s^{-2}. [2]

(b) Answer: _____ N

1.6 Fairground ride A.2 · Circular motion · [2 marks]

A seat on a fairground ride moves in a horizontal circle of radius 9.0 m and completes one revolution every 7.0 s.

(a) Calculate the speed of the seat. [1]

(a) Answer: _____ $m\ s^{-1}$

(b) Calculate the centripetal acceleration. [1]

(b) Answer: _____ $m\ s^{-2}$

1.7 Goalkeeper's save A.2 · Momentum and impulse · [2 marks]

A goalkeeper stops a 0.43 kg football travelling at 17 $m\ s^{-1}$. The ball comes to rest in 0.12 s. Calculate the average force on the ball.

Answer: _____ N

1.8 Trolleys couple together A.2 · Momentum and impulse · [2 marks]

A 1.0 kg trolley moving at 0.60 $m\ s^{-1}$ collides with and couples to a 0.30 kg trolley at rest. Calculate their common velocity.

Answer: _____ $m\ s^{-1}$

Level 2 · Multi-step — two or three linked steps

8 exercises · 32 marks · show your working

2.1 Sledge pulled at an angle A.2 · Newton's laws · [5 marks]

A 32 kg sledge is pulled across level snow by a rope at 35° above the horizontal with a force of 130 N. The coefficient of dynamic friction is 0.17.

(a) Calculate the normal force on the sledge. [2]

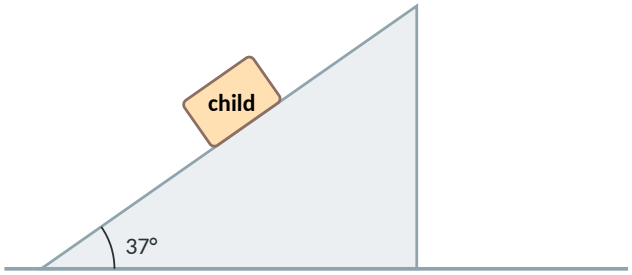
(a) Answer: _____ N

(b) Calculate the acceleration of the sledge. [3]

(b) Answer: _____ m s^{-2}

2.2 Playground slide A.2 · Newton's laws · [4 marks]

A child slides from rest down a 5.0 m long slide inclined at 37° to the horizontal. The coefficient of dynamic friction is 0.27.



(a) Calculate the acceleration of the child. [3]

(a) Answer: _____ m s^{-2}

(b) Calculate the speed of the child at the bottom. [1]

(b) Answer: _____ m s^{-1}

2.3 Two blocks on a string A.2 · Newton's laws · [4 marks]

Blocks of 6.0 kg and 4.0 kg on a frictionless table are joined by a light string. A horizontal force of 26 N pulls the 4.0 kg block; the 6.0 kg block trails behind.

(a) Calculate the acceleration of the blocks. [2]

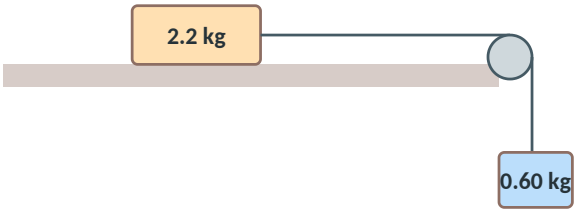
(a) Answer: _____ m s^{-2}

(b) Calculate the tension in the string. [2]

(b) Answer: _____ N

2.4 Cart pulled by a hanging mass A.2 · Newton's laws · [4 marks]

A 2.2 kg cart on a smooth horizontal table is attached by a light string over a smooth pulley to a hanging mass of 0.60 kg. The system is released from rest.



(a) Calculate the acceleration of the system. [2]

(a) Answer: _____ m s^{-2}

(b) Calculate the tension in the string. [1]

(b) Answer: _____ N

(c) Calculate the time for the cart to move 1.0 m from rest. [1]

(c) Answer: _____ s

2.5 Bathroom scale in a lift A.2 · Newton's laws · [4 marks]

A 72 kg student stands on bathroom scales (which measure the normal force) in a lift.

(a) Calculate the scale reading when the lift accelerates upwards at 1.7 m s^{-2}. [2]

(a) Answer: _____ N

(b) Calculate the scale reading when the lift is moving upwards but slowing down at 1.7 m s^{-2}. [2]

(b) Answer: _____ N

2.6 Speed limit on a flat bend A.2 · Circular motion · [4 marks]

A car drives round a flat bend of radius 90 m. The coefficient of static friction between the tyres and the road is 0.80.

(a) Calculate the maximum speed at which the car can go round the bend without skidding. [3]

(a) Answer: _____ m s^{-1}

(b) Convert this speed to km h^{-1}. [1]

(b) Answer: _____ km h^{-1}

2.7 Ball rebounds from a wall A.2 · Momentum and impulse · [3 marks]

A 0.60 kg ball hits a wall at 7.0 m s^{-1} and rebounds along the same line at 5.0 m s^{-1}. It is in contact with the wall for 0.060 s.

(a) Calculate the magnitude of the change in momentum of the ball. [2]

(a) Answer: _____ kg m s^{-1}

(b) Calculate the average force exerted by the wall on the ball. [1]

(b) Answer: _____ N

2.8 Hump-backed bridge A.2 · Circular motion · [4 marks]

A 1100 kg car drives over a hump-backed bridge whose top is part of a vertical circle of radius 22 m.

(a) Calculate the normal force on the car at the top of the bridge when its speed is 12 m s^{-1}. [3]

(a) Answer: _____ N

(b) Calculate the speed at which the car would just lose contact with the road. [1]

(b) Answer: _____ m s^{-1}

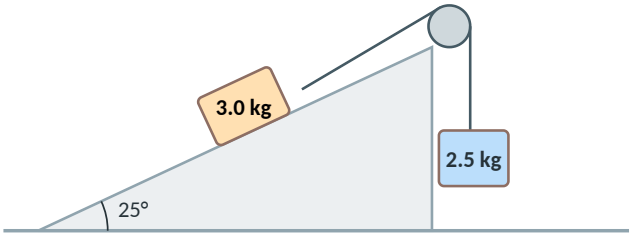
Level 3 · Paper 2 structured — parts and marks

8 exercises · 65 marks · show your working

3.1 Ramp, pulley and hanging mass

A.2 · Newton's laws · [10 marks]

A 3.0 kg block rests on a ramp inclined at 25°. It is connected by a light string over a smooth pulley at the top of the ramp to a hanging mass of 2.5 kg. Between the block and the ramp $\mu_s = 0.40$ and $\mu_d = 0.25$.



(a) Calculate the component of the block's weight parallel to the ramp. [1]

(a) Answer: _____ N

(b) Calculate the maximum static friction force on the block. [2]

(b) Answer: _____ N

(c) Determine whether the block starts to move when the system is released. Explain. [2]

(d) Calculate the acceleration of the system once it is moving. [3]

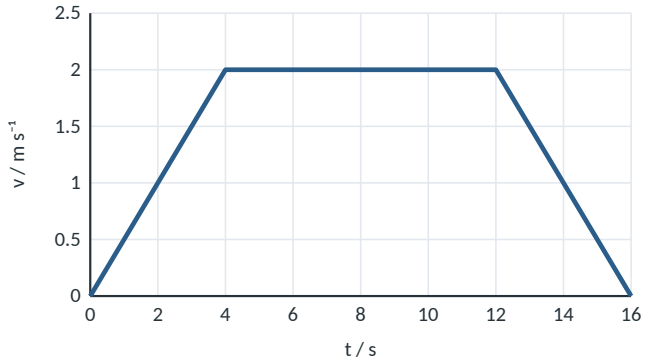
(d) Answer: _____ $m\,s^{-2}$

(e) Calculate the tension in the string. [2]

(e) Answer: _____ N

3.2 Crane lifting a load A.2 · Newton's laws · [8 marks]

A crane lifts a 600 kg load vertically. The graph shows the velocity of the load against time.



(a) Calculate the tension in the cable between t = 0 and t = 4.0 s. [2]

(a) Answer: _____ N

(b) Calculate the tension between t = 4.0 s and t = 12 s. [1]

(b) Answer: _____ N

(c) Calculate the tension between t = 12 s and t = 16 s. [2]

(c) Answer: _____ N

(d) Calculate the height through which the load is lifted. [2]

(d) Answer: _____ m

(e) Explain why the cable is most likely to break during the first 4.0 s. [1]

3.3 Bullet into a block A.2 · Momentum and impulse · [8 marks]

A 15 g bullet travelling horizontally at 320 m s^{-1} embeds itself in a 2.5 kg wooden block at rest on a rough horizontal floor. The coefficient of dynamic friction between the block and the floor is 0.40.

(a) Calculate the speed of the block (with the bullet) just after the impact. [2]

(a) Answer: _____ m s^{-1}

(b) Calculate the friction force on the block as it slides. [1]

(b) Answer: _____ N

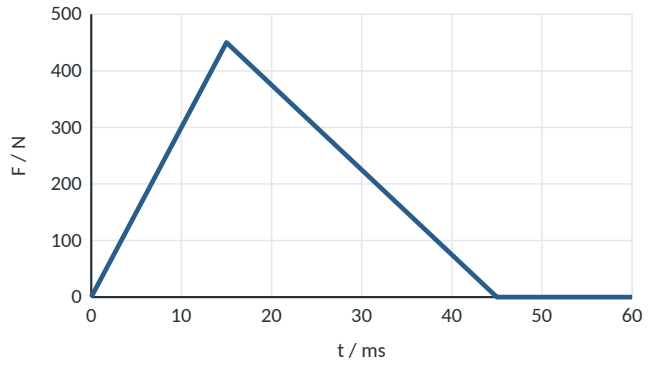
(c) Calculate the distance the block slides before stopping. [3]

(c) Answer: _____ m

(d) Explain why momentum can be taken as conserved during the impact even though friction acts on the block. [2]

3.4 Baseball and bat A.2 · Momentum and impulse · [7 marks]

A 0.145 kg baseball moving at 38 m s^{-1} towards a bat is hit straight back. The graph shows how the force from the bat varies with time.



(a) Calculate the impulse given to the ball. [2]

(a) Answer: _____ N s

(b) Calculate the speed of the ball as it leaves the bat. [3]

(b) Answer: _____ m s^{-1}

(c) Calculate the average force on the ball during the contact. [1]

(c) Answer: _____ N

(d) Calculate the maximum acceleration of the ball. [1]

(d) Answer: _____ m s^{-2}

3.5 Underwater robot A.2 · Types of forces · [9 marks]

An underwater robot (ROV) of mass 150 kg and volume 0.140 m^3 descends vertically through seawater (ρ = 1025 kg m^{-3}) at constant velocity with its motors off.

(a) Calculate the weight of the ROV. [1]

(a) Answer: _____ N

(b) Calculate the buoyant force on the ROV. [1]

(b) Answer: _____ N

(c) Calculate the drag force on the ROV. [2]

(c) Answer: _____ N

(d) The ROV now releases 15 kg of ballast (of negligible volume). Calculate its acceleration immediately afterwards. [3]

(d) Answer: _____ $m\ s^{-2}$

(e) Describe the motion of the ROV after the ballast is released. [2]

3.6 Spring data A.2 · Types of forces · [7 marks]

A student hangs masses from a spring and measures its extension:

m / kg	0.10	0.20	0.30	0.40	0.50
Δx / cm	3.1	6.1	9.2	12.3	15.3

(a) Calculate the force exerted by the spring on the 0.30 kg mass. [1]

(a) Answer: _____ N

(b) Determine the spring constant, using the first and last data points. [2]

(b) Answer: _____ $N\ m^{-1}$

(c) Calculate the extension when a 0.75 kg mass hangs at rest from the spring (assume it obeys Hooke's law). [1]

(c) Answer: _____ m

(d) The spring with a 0.40 kg mass is hung in a lift accelerating upwards at 1.5 $m\ s^{-2}$. Calculate the extension. [2]

(d) Answer: _____ m

(e) State one reason why the prediction in (c) might be wrong. [1]

3.7 Car skids off a bend A.2 · Circular motion · [8 marks]

A car skidded off a flat bend of radius 50 m. Tests show $\mu_s = 0.65$ between the tyres and the dry road. The speed limit is 50

km h^{-1}.

(a) Calculate the maximum speed at which a car can take this bend on a dry road. [2]

(a) Answer: _____ m s^{-1}

(b) Convert your answer to km h^{-1}. [1]

(b) Answer: _____ km h^{-1}

(c) On the day of the accident the road was wet (μ_s = 0.30). Calculate the maximum safe speed in km h^{-1}. [2]

(c) Answer: _____ km h^{-1}

(d) The driver says she kept to the speed limit. Discuss whether this means she drove safely, and describe the path of the car once it started to skid. [2]

(e) Explain why the answer does not depend on the mass of the car. [1]

3.8 Rear-end collision A.2 · Momentum and impulse · [8 marks]

A 1200 kg car travelling at 12.5 m s^{-1} runs into the back of a stationary 1600 kg car. The cars lock together. The collision lasts 0.15 s.

(a) Calculate the velocity of the cars just after the collision. [2]

(a) Answer: _____ m s^{-1}

(b) Calculate the impulse on the stationary car. [1]

(b) Answer: _____ N s

(c) Calculate the average force between the cars. [1]

(c) Answer: _____ N

(d) A 70 kg driver in the moving car is held by a seat belt. Calculate the average deceleration of the driver as a multiple of g. [2]

(d) Answer: _____

(e) Explain how a crumple zone reduces the force on the driver. [2]

End of exercises